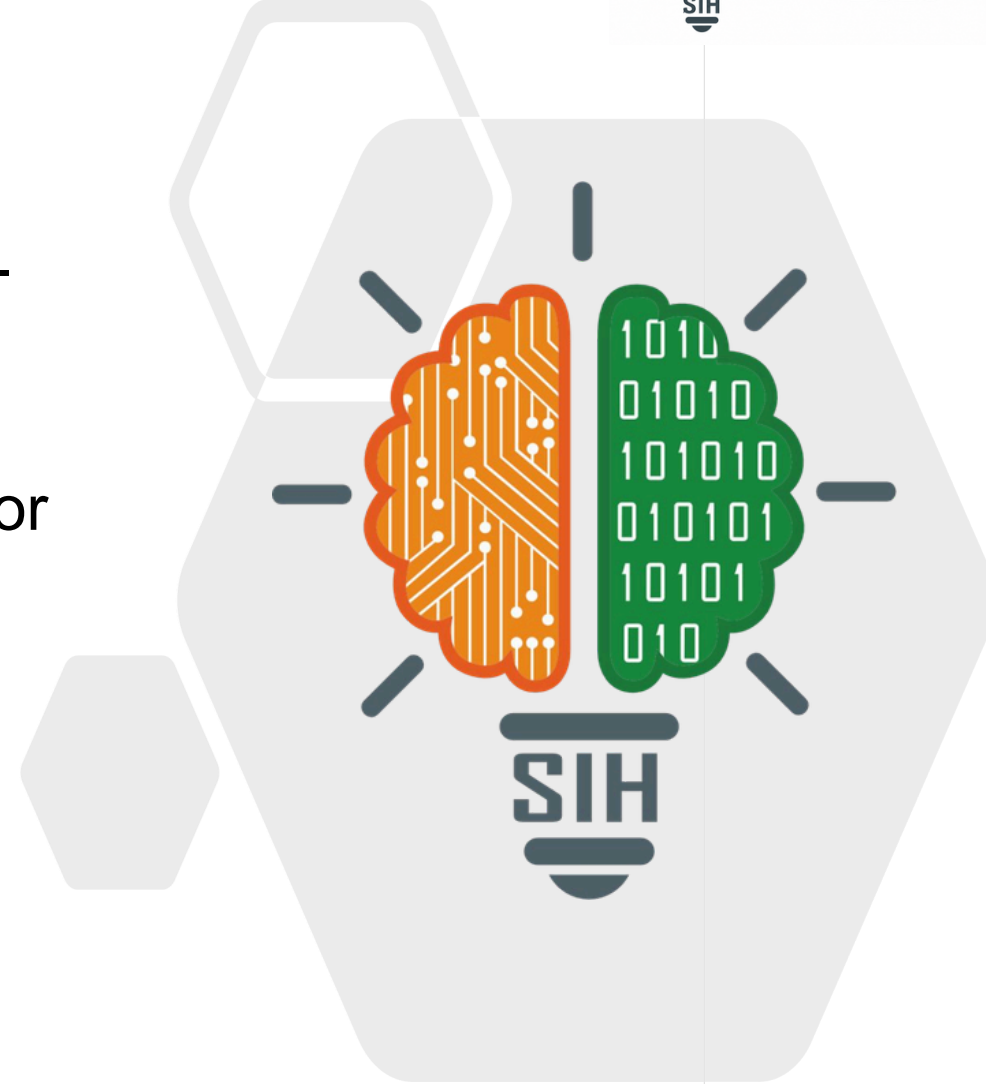


SMART INDIA HACKATHON 2026



- **Problem Statement ID** - SIH26025
- **Problem Statement Title** - Development of an AI-enabled Low Cost Real Time Mine Subsidence Monitoring, Prediction and Early Warning System for Underground Coal Mines in India
- **Theme** - Smart Automation
- **PS Category** - Hardware
- **Team ID**- 106
- **Team Name** - MineWatch



IDEA/ SOLUTION

- Our proposed solution uses **smart sensor nodes** deployed above underground mining areas, integrating a geophone for vibration, an FBG sensor for crack/displacement, and an IMU for abnormal movement and tilt monitoring.
- Sensor data is processed by an STM32 microcontroller, which detects significant changes, generates data packets, and transmits them through a 915 MHz LoRa network from node to node to the central [dashboard](#).
- Each node uses a **solar panel with battery backup**, designed to operate for approximately **one year without solar input**.
- During LoRa failure, the STM32 activates external FRAM (Non-Volatile Memory) to temporarily store sensor data while monitoring continues. Only the initial reading and subsequent changed values are stored, with **communication checked every five minutes**.
- Once communication is restored, the **buffered data** is transmitted through the LoRa network and normal real-time transmission resumes.
- The central dashboard provides **live node locations**, node conditions, self-diagnostics, LoRa status and alerts, transmission status, and real-time/historical sensor data.
- Mining activity, surface conditions, rainfall and environmental factors are incorporated to distinguish operational or natural disturbances from potentially significant mining-related anomalies.
- Initially, an Isolation Forest model detects unusual patterns in vibration, displacement, cracks, tilt and movement without requiring labelled data.
- Approximately six months of collected field data will progressively improve the ML model, enabling more refined site-specific **subsidence monitoring and early-warning analysis**.
- **Cross-sensor validation** compares vibration, tilt, displacement and strain readings from nearby nodes to confirm whether an observed change represents actual ground movement rather than a single-sensor fault or local disturbance, reducing false alarms.

PROBLEM RESOLUTION

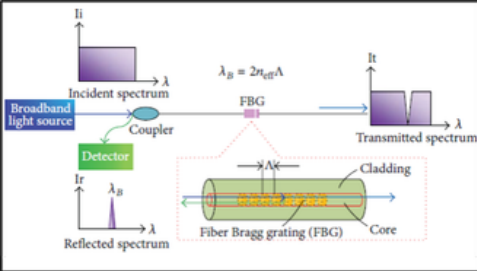
- **Context-Aware AI Detection:** **AI-based anomaly detection** combines sensor data with mining activity, surface conditions, rainfall, and environmental factors to distinguish operational or natural disturbances from potential subsidence.
- **Scalable IoT Network:** STM32-LoRa sensor nodes provide **low-power, long-range communication** with **solar and battery backup**, while **FRAM** stores changed data during communication failures.
- **GIS-Based Monitoring:** An interactive dashboard displays live node locations, sensor conditions, historical trends, communication status, and automated anomaly alerts.
- **Cross-Sensor Validation:** Data from multiple nearby sensors is compared to distinguish genuine ground movement from isolated sensor faults or local disturbances, reducing false alarms.

UNIQUE VALUE PROPOSITION

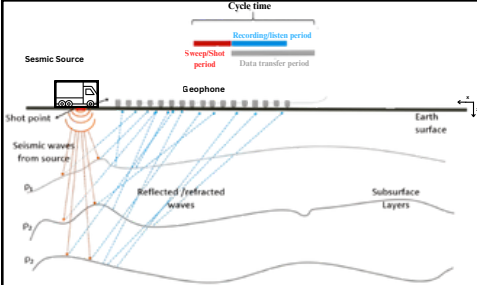
- **Modified FBG-based sensing system for surface deformation**
- **12-months battery life without solar power**
- **Robust, low power, scalable, and suitable for Indian geo-mining conditions**
- **AI/ML-based risk prediction using Isolation Forest for unlabeled data, classifying areas into Safe, Moderate, and Unsafe zones.**
- **Coupler-to-coupler design for easy replacement.**
- **Operator-directed early warning alerts.**
- **Cloud-synchronized historical + real-time data**
- **Self-diagnostic system – sensor, battery & communication health**
- **Multilingual dashboard interface.**
- **Intelligent F-RAM data buffering**



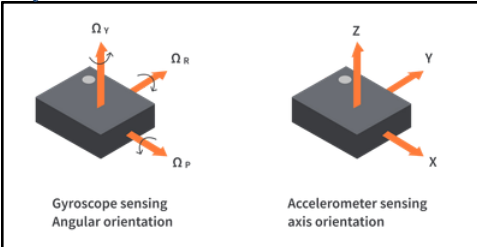
Modified FBG



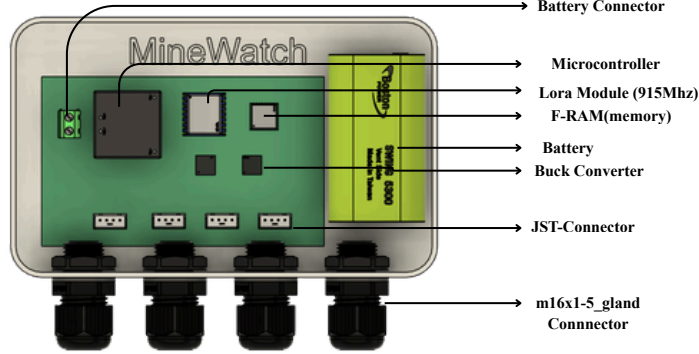
Geophone



Gyro/Accelerometer



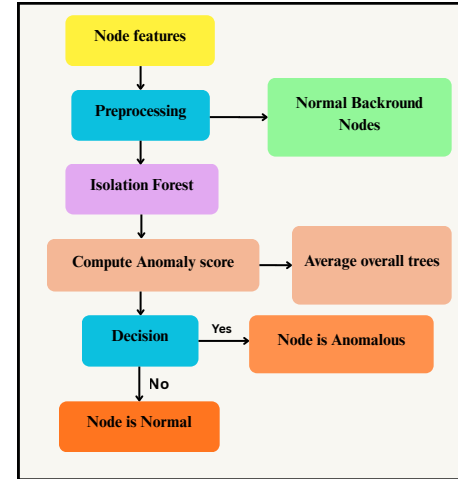
internal componet picture



DEPLOYMENT GUIDE

- **Site survey:** The **panel boundary** is taken from the survey plan and the grid origin is fixed at a survey monument.
- **Grid alignment:** The grid is aligned to the strike and dip axes so that nodes follow the direction of ground movement.
- **Spacing selection:** Nodes are spaced 50 m apart in normal-risk areas (4 nodes/ha) and 25 m apart near goaf or fault zones (16 nodes/ha).
- **Node count:** Nodes per panel = $(L/50 + 1) \times (W/50 + 1)$, so a 500×300 m panel needs $11 \times 7 = 77$ nodes.
- **Gateway positioning:** The gateway is placed at the **grid centre**. Its 500 m range covers $\pi \times 500^2 \approx 78.5$ ha, and the link is tested with RSSI/SNR.
- **Verification and finalization:** **Planned positions** are staked out and surveyed with GNSS/RTK (± 3 m / ± 0.3 m), and the as-built coordinates are recorded in the GIS.

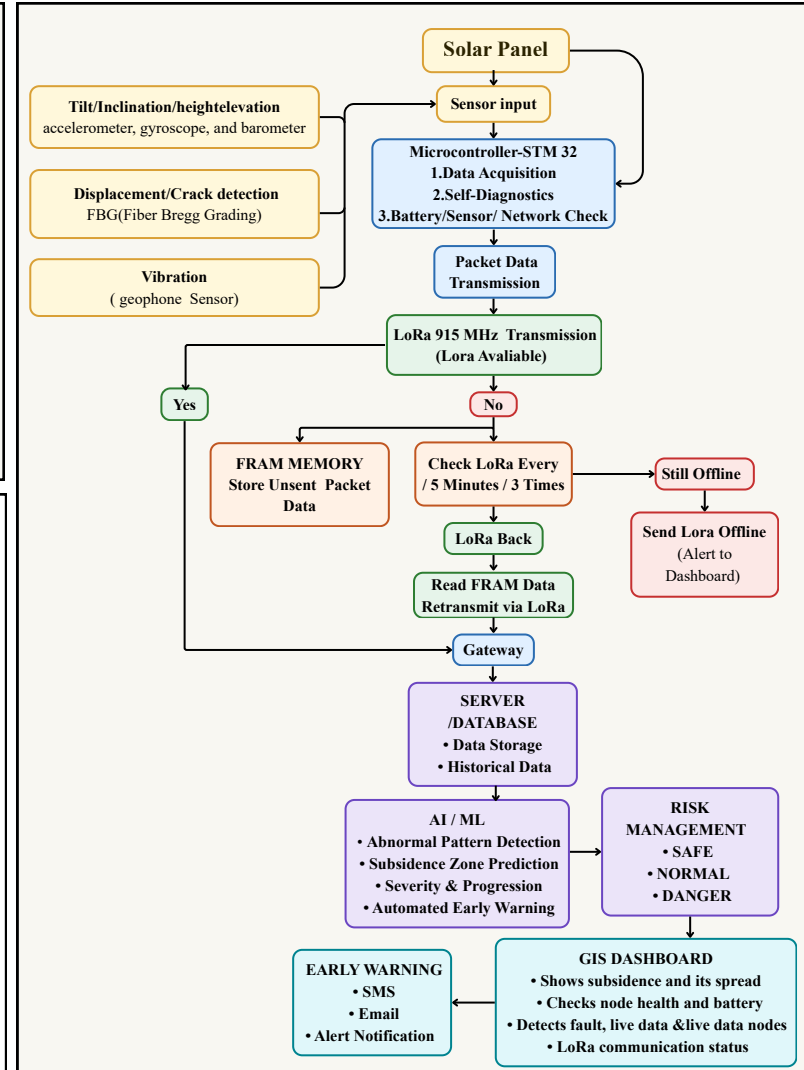
ALGORITHM WORKFLOW



Material Required

- **STM32 MCU**-Collects and processes sensor data.
- **FBG Sensor**-Detect Crack & deformation.
- **Geophone**-Detect Ground vibration.
- **Accelerometer**-Detects ground movement and acceleration.
- **Gyroscope**-Detects ground **tilt** and angular movement.
- **Barometer**-Measures air pressure.
- **LoRa 915 MHz**-Wireless data transmission over **long distance**.
- **FRAM**-Stores data when LoRa communication fails.
- **Solar Panel + Battery**- Solar panel generates electricity, while the battery stores energy to power the system continuously.

SYSTEM WORKFLOW



ANALYSIS OF FEASIBILITY

1. Technical & Sensor Feasibility

- **Multi Parameter Sensing:** A **geophone** monitors **ground vibration**, an **FBG sensor** detects **crack initiation and displacement**, and an IMU (accelerometer + gyroscope) **tracks tilt/inclination and abnormal movement**.
- **Edge Processing:** An STM32 microcontroller **processes sensor signals**, identifies abnormal changes and **converts them into data packets**.
- **Wireless Mesh Communication:** Packets hop from **node to node** over LoRa 915 MHz until they reach the central dashboard.
- **Energy Autonomy:** Each node runs on a **solar power panel with battery backup** and can operate for a out one year without solar power

2. Economic & Scalability Feasibility

- **Low-Cost Distributed Monitoring:** **Geophone** and **IMU** nodes are inexpensive to deploy widely, while FBG sensors are placed selectively at critical locations.
- **Modular Node Design:** Every node uses the **same hardware**, so coverage grows by **simply adding nodes to the LoRa mesh**.
- **Infrastructure-Free Coverage:** Multi-hop LoRa links **extend range** across the surface without cabling or extra gateways.
- **Low-Power Operation:** **Change-based data storage** and efficient communication cycles **reduce energy use** and memory wear.

3. Operational & Software Feasibility

- **Fault-Tolerant Data Handelling:** On a LoRa failure, the node buffers data in **FRAM**, **stores only the first value** and later changed values and **checks the link every 5 minutes**. When the link returns, **the sends the buffered data and marks it as transmitted**.
- **Centralized Dashboard:** **Multilingual access**, live node locations, node health and self diagnostics, LoRa status and failure alerts, transmission status and data quantity and **real-time plus historical sensor data**, all saved in a secure backend database.
- **Phased AI Analysis:** Isolation Forest anomaly detection **works from initially with no labelled dataset**, model **is retrained for data-driven mine-subsidence prediction after 6 months**

POTENTIAL CHALLENGES, RISKS AND STRATEGIES FOR OVERCOMING THESE CHALLENGES

Aspect / Property	Challenges	Risks	Strategies to Overcome	Conventional	Proposed System
MCU / Controller	Multi-Sensor	Data Loss	FRAM	Basic Mcu	STM32F401RE
Displacement / Stretch	FBG coupling	Deformation error	Calibration + Compensation	Point measurement	Continuos FBG crack & displacement detection
Vibration	Background noise	False events	Noise filtering + Sensor fusion	Manual / periodic checks	Continuous geophone monitoring
Sensor Reliability	Drift / Failure	False alert	Health + Node validation	Manual verification	Self diagnosing nodes
Power Supply	No grid, low sunlight	Node shutdown	Solar + Battery (~1 yr backup)	Wired / manual power	Self-powered nodes
Risk Prediction	No labled dataset	Prediction uncertainty	Isolation Forest → ML retraining after ~6 months	Reactive assessment	Phased AI early warning
Connectivity	LoRa link failure	Data loss / Delay	Multi-hop mesh + FRAM buffering	Connectivity dependent	Offline-capable, auto-resend
Tilt / Inclination	Micro-angle changes	Drift / False alarm	Calibration + AI filtering	Periodic survey	Continuous IMU tilt monitoring

POTENTIAL IMPACT ON THE TARGET AUDIENCE

1. Mine Operators and Management

- **Risk Prevention:** **Real-time monitoring** helps identify early signs of subsidence.
- **Operational Planning:** Ground-condition data supports **inspection and maintenance decisions**.
- **Reduced Survey Effort:** **Continuous sensing** reduces dependence on frequent manual surveys.
- **Business Continuity:** Early warning enables **timely action** and helps prevent unexpected **mine shutdowns and operational delays**.

2. Mine Workers and Nearby Communities

- **Worker Protection:** **Early-warning alerts** provide timely information about **abnormal ground conditions**.
- **Emergency Awareness:** **Real-time alerts** enable quick response to **hazardous ground conditions**.
- **Public Awareness:** **GIS-based visualization** helps classify areas into **safe and danger zones** based on real-time and historical sensor data.
- **Asset Protection:** Early detection of **ground movement and cracks** helps protect infrastructure, agricultural land and nearby communities.

3. Government, Regulators and Mine Safety Agencies

- **Digital Monitoring:** Continuous sensor data provides a **digital record** of ground conditions.
- **Centralized Oversight:** The dashboard enables **remote monitoring** of multiple sensor nodes.
- **Data-Driven Assessment:** **Historical data** supports mine-safety assessment and planning.
- **Scalable Monitoring:** The system can be expanded across **multiple underground coalfields**.

4. Environmental Stakeholders

- **Land Conservation:** Monitoring helps identify vulnerable surface areas.
- **Damage Mitigation:** **Early detection** of deformation can support protection of land and infrastructure.
- **Responsible Mining:** Continuous monitoring supports safer and sustainable mining practices.
- **Ground Behaviour Studies:** Long-term data helps analyse **subsidence patterns and progression**.

5. Engineering and Technology Sector

- **Indigenous Solution:** Develops a low-cost, Made-in-India mining system.
- **Smart Technology:** Combines IoT, AI/ML, LoRa and GIS technologies.
- **Scalable Network:** Supports **low-cost** smart sensor nodes across mine areas.
- **Student-Friendly:** Uses readily available hardware for easy development and deployment.

BENEFITS OF THE SOLUTION

1. Safety Benefits

- **Early Detection:** **Real time continuous monitoring** identifies ground movement, tilt and cracks at an early stage.
- **Worker Safety:** **Early-warning alerts** support timely safety action.
- **Risk Reduction:** Helps **identify potential danger zones** before conditions worsen.
- **Reliable Monitoring:** **Battery backup** supports continuous monitoring during power interruptions.

2. Economic Benefits

- **Lower Cost:** Low-cost STM32 and LoRa nodes **reduce monitoring costs**.
- **Damage Reduction:** **Early detection** can help reduce subsidence-related damage to infrastructure.
- **Low-Power Operation:** **Solar + battery** reduces regular power requirements.
- **Scalable Deployment:** **Additional modular sensor nodes** can be added across a large mining area and multiple underground panels.

3. Operational Benefits

- **Real-Time Monitoring:** Central dashboard provides **continuous live ground-condition data** for a faster assessment and decision making.
- **AI-Based Detection:** Identifies **abnormal sensor patterns** using AI/ML and display on dashboard.
- **Better Decision Making:** GIS visualization helps **identify risk areas**.
- **Self-Diagnostics:** STM32 Monitors sensors, battery, node and LoRa status and report node health to dashboard.

4. Technological Benefits

- **Multi-Parameter Sensing:** Combines FBG, geophone, accelerometer, gyroscope and barometer.
- **Long-Range Communication:** LoRa 915 MHz provides **low-power wireless communication**.
- **FRAM Data Buffering:** Stores important sensor data during LoRa failure.
- **Smart Data Storage:** Stores the first value and meaningful changes, reducing unnecessary memory use.
- **Progressive AI:** Uses collected **real-world data** to improve future prediction models.
- **Integrated Platform:** Combines IoT, AI/ML, LoRa, FRAM, GIS and dashboard.

RESEARCH	REFERENCES
DOCX	link
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Earthformer_MSTSP: A Dynamic Simulation-Based Model for Mining Subsidence Prediction Using InSAR Observations	https://ieeexplore.ieee.org/document/11247827?denied=
Real-Time Monitoring of Underground Mines with IoT and Machine Learning Integration	https://informaticsjournals.co.in/index.php/jmmf/article/view/48456
An Overview of GIS-Based Assessment and Mapping of Mining-Induced Subsidence	https://www.mdpi.com/2076-3417/10/21/7845
Automatic crack detection method for loaded coal in vibration failure process	https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0185750